

ROY OKOLA OTIENO

Technical Sales & Operations Engineer | Full-Stack Engineer | Clean Energy & EV Infrastructure

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PROFILE

Technical Operations & Sales Engineer with 3+ years driving Kenya's solar PV and EV charging sectors — leading multi-site feasibility studies, B2B key account sales, EPRA/EMAK regulatory engagement, and technical training for Roam Electric's growing portfolio. Independent full-stack engineer at rauell.systems, shipping live products across energy, finance, and WASH sectors. Africa Fellow, Young Energy Leaders Cohort 5 (Completed 2026). MBA candidate, University of East London.

CORE SKILLS & COMPETENCIES

Energy & EV: Solar PV Design · Solar Hybrid Sizing · EV Charging Infrastructure · MPPT · Deye · Solis · Jinko Solar · Battery Storage · PVsyst · PV*SOL · HOMER

Technical Ops: Multi-Site Operations · Technical Feasibility Studies · KPI Reporting · Fault Escalation · Site Commissioning · EPRA/EMAK Regulatory Compliance

Sales & BD: B2B Key Account Management · Technical Proposals · CRM · Stakeholder Mapping · Client Training · Lead Conversion

Engineering: Full-Stack Development · Next.js · React · Node.js · Supabase · PostgreSQL · Vercel · REST APIs · M-Pesa Integration

PROFESSIONAL EXPERIENCE

Technical Operations & Sales Engineer

Jun 2025 – Present

Roam Electric Limited · Nairobi, Kenya · Hybrid

- Conduct multi-site technical assessments and feasibility studies for EV charging and solar hybrid system deployments across Nairobi and Kiambu — coordinating field technicians, clients, and internal engineering teams end-to-end.
- Monitor energy system performance across a 25%+ expanded portfolio of active EV and solar sites; identify and escalate technical faults to maintain uptime targets.
- Size and quote solar hybrid systems using Deye inverters, Solis string inverters, and Jinko solar panels — proposals include MPPT voltage verification and battery bank sizing.
- Deliver 10+ product and technical trainings to commercial and fleet clients on EV hardware, charging systems, and energy storage components.
- Support EV charging network expansion across Kenya including a 7-station rollout with Be Energy Limited — managing site assessments, partner coordination, and commercial tenancy negotiations.

Founder & Full-Stack Engineer

Jan 2024 – Present

rauell.systems · Nairobi, Kenya · Remote

- SafariCharge: Solar monitoring and financial simulation dashboard for the Kenyan market (React, Next.js, Vercel).
- finance.rauell.systems: Personal finance tracker with M-Pesa SMS automation pipeline (MacroDroid + Supabase + Node.js) — fully deployed and in production.
- WASH Governance Platform (kypw.org): Reporting and coordination tool for Kenya Youth Parliament for Water (Next.js, Supabase, PostgreSQL).
- Roam Electric Charging Points Tracker: Real-time EV charging infrastructure visibility platform for Kenya (Next.js 16, Neon PostgreSQL, NextAuth) — live at charging-points-iota.vercel.app.

Portfolio: rauell.systems · Stack: Next.js · React · Node.js · Supabase · PostgreSQL · Vercel · Upstash

Infrastructure Engineer

Apr 2025 – Jun 2025

EVChaja · Nairobi, Kenya · Hybrid

- Led technical scoping and infrastructure analysis for 3 investment-ready EV charging projects — engineering reports supported stakeholder consultations that attracted KES 50M+ in potential investment.
- Engaged EPRA and EMAK on regulatory compliance and technical risk mapping across charging hardware, power electronics, battery management systems (BMS), and fleet-based energy models.

Technical Sales Engineer Intern

Feb 2024 – Jul 2024

Frisco Engineering Limited · Utawala, Kenya · On-site

- Designed and delivered 10+ solar PV and backup power systems (KES 3M+ total project value) — conducting 15+ energy audits and site feasibility assessments for agricultural and commercial clients.
- Used PVsyst and PV*SOL for solar system modelling; prepared SLDs, BOMs, and technical proposals for off-grid and grid-tied systems.
- Managed system commissioning, client handover, and after-sales technical support — achieving 95% customer satisfaction through structured issue tracking and resolution.

LEADERSHIP & COMMUNITY ENGAGEMENT

- Africa Fellow, Young Energy Leaders — Cohort 5 (Completed 2026). Multidisciplinary fellowship advancing clean-energy innovation and the African energy transition; Solar PV Engineering Track completed Jan 2026.
- Event Coordinator & Member, Kenya Youth Parliament for Water (KYPW) — Lead webinars, policy dialogues, and field outreach for the national chapter. Coordinated Menstrual Hygiene Day 2026 at Lang'ata Women's Prison, featured in Citizen Digital.
- Community Trainer — Workshops on solar irrigation, renewable energy, and circular-economy practices for rural farmers.

EDUCATION

Master of Business Administration (MBA)

2025 – 2027 (Expected)

University of East London · via Unicaf · London, UK

B.Sc. Agricultural & Biosystems Engineering

Sep 2018 – Dec 2023

Jomo Kenyatta University of Agriculture and Technology (JKUAT) · Juja, Kenya · Second Class Honours, Upper Division

Projects: Designed a solar-powered evaporative cooling system extending tomato shelf life by 7 days in off-grid areas; contributed to feasibility studies for 10+ EV charging hubs integrating solar and second-life batteries.

LICENCES & CERTIFICATIONS

Battery Technology Training — Advanced	Siemens Stiftung / pManifold EV Academy	Oct 2025
Battery Technology Training — Basic	Siemens Stiftung / pManifold EV Academy	Sep 2025
Elum Certified Installer — ePowerControl ES/MC Series	Elum Energy	Dec 2025
Electric Mobility Technical Training	Advanced Mobility Centre / Kenya Power	Jan 2025
Africa Fellowship — Solar PV Engineering Track, Cohort 5	Africa Fellowship for Young Energy Leaders	Jan 2026
Solar Systems Design, Sizing & Commissioning (T2)	Solar Training Kenya	Feb 2025
Data Fundamentals for Sustainable Mobility	TUMI / GIZ / Cities Forum	2025
Electric Two- and Three-Wheelers: Steering the Mobility Revolution	TUMI / PEM Motion	Jun 2024
Exploring the World of Electric Buses	TUMI / PEM Motion	Jun 2024
Electric Mobility: Key Concepts & Strategies	TUMI / PEM Motion	Nov 2025